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Education

- 2024 – 2025 **M.Sc. in Electronics and Communication Engineering, Khulna University.**
CGPA: 3.88/4.00, Total Credit: 40
Thesis title: *Integrated Surveillance System for Khulna University: Enhancing Safety, Security, and Sustainability through Camera Footage and Peripheral Sensors.*
- 2019–2023 **B.Sc. in Electronics and Communication Engineering, Khulna University.**
Four Year Honours Degree, CGPA: 3.59/4.00 (75.80% marks), Total Credits: 160.
Thesis title: *Multi-Platform Data Security Systems using n-Bit Modified (n-BM) and Reusable Block (n-RB) AES.*

Research Publications

Journal Articles

- 1 P. Sarker, A. Ksibi, M. M. Jamjoom, K. Choi, A. A. Nahid, and M. A. Samad, "Breast cancer prediction with feature-selected xgb classifier, optimized by metaheuristic algorithms," *Journal of Big Data*, vol. 12, no. 1, p. 78, 2025.
- 2 P. Sarker, K. Choi, A.-A. Nahid, and M. A. Samad, "Catboost with physics-based metaheuristics for thyroid cancer recurrence prediction," *BioData Mining*, vol. 18, no. 1, p. 84, 2025.
- 3 P. Sarker, J.-J. Tiang, and A.-A. Nahid, "Metaheuristic-driven feature selection for human activity recognition on ku-har dataset using xgboost classifier," *Sensors*, vol. 25, no. 17, p. 5303, 2025.
- 4 P. Sarker, J.-J. Tiang, and A.-A. Nahid, "Gallstone classification using random forest optimized by sand cat swarm optimization algorithm with shap and dice-based interpretability," *Sensors*, vol. 25, no. 17, p. 5489, 2025.
- 5 P. Sarker, J.-J. Tiang, and A.-A. Nahid, "Dengue fever detection using swarm intelligence and xgboost classifier: An interpretable approach with shap and dice," *Information*, vol. 16, no. 9, p. 789, 2025.
- 6 P. Das, P. Sarker, J.-J. Tiang, and A.-A. Nahid, "Breast cancer prediction using rotation forest algorithm along with finding the influential causes," *Bioengineering*, vol. 12, no. 10, p. 1020, 2025.
- 7 P. Das, P. Sarker, J.-J. Tiang, and A.-A. Nahid, "Dengue fever classification integrating bird swarm algorithm with gradient boosting classifier along with feature selection and shap–dice based interpretability," *Applied Sciences*, vol. 15, no. 21, p. 11 413, 2025.
- 8 K. Shrestha, P. Sarker, J.-J. Tiang, and A.-A. Nahid, "Metaheuristic-based gallstone classification using rotational forest explained with shap," *Frontiers in Digital Health*, vol. 7, p. 1 727 559, 2025.

Conference Proceedings

- 1 M. I. H. Sikder, P. Sarker, M. T. Hasan, and M. I. Kadir, "Reusable block aes: A modified aes algorithm for the encryption and decryption of sensitive data," in *2025 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, IEEE, 2025, pp. 1–6.

- 2 M. I. H. Sikder, P. Sarker, and M. I. Kadir, "A smartphone application for multiformat data security using n-bit reusable block aes," in *2025 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, IEEE, 2025, pp. 1–6.

Research Under Progress

1. Feature Reduction Using Swarm Optimization and Random Forest Classifiers for Early Diabetes Risk Prediction, *Scientific Reports* (**Accepted**)
2. Gradient Boosting Classifier with Zebra Optimization Algorithm for Pregnancy Risk Prediction, *ICT Express* (**Under Review**)
3. Swarm-Based Metaheuristic Algorithms and Classifiers for Predicting Myocardial Infarction, *The Journal of Supercomputing* (**Under Review**)
4. Swarm-Optimized Machine Learning for Rapid and Explainable Diagnosis of Rheumatic and Autoimmune Diseases, *Artificial Intelligence In Medicine* (**With Editor**)

Projects

App for Data Encryption	■	Developed an APK to securely encrypt images and text; part of my undergraduate thesis.
Height Measurement Machine	■	Arduino UNO–based non-contact height measurement using ultrasonic sensor and Android app.
CRC Simulator	■	Simulator for binary data transmission; indicates errors via syndromes.

Skills

Languages	■	Strong reading, writing and speaking competencies for English and Bangla.
Coding	■	Python (Classification; Metaheuristic Algorithms: SLO, WOA, FOX, BESA, EOA, EVOA, HBA, SCSO, and others; XAI: SHAP and LIME, and DiCE), Matlab, C/C++, JAVA, $\LaTeX$.
Software	■	MS Office (Word, Excel, PowerPoint, Visio); basic Android Studio and Cisco Packet Tracer.
Embedded Systems	■	Arduino, NVIDIA Jetson Developer Kit.
Others	■	Academic research, journal writing, and $\LaTeX$ typesetting using Overleaf and TeXstudio, with several publications.

Experience

Research Assistant

2025 ■ Developing Security System using Jetson Developer Kit, YOLO, Viola Jones, and ViT Algorithms.

Industrial Training

2023 ■ Khulna 225 MW Combined Cycle Power Plant.

Certification/Award

2024 ■ **Certificate of Appreciation.** IEEE SPICSCON 2024, for outstanding contributions as Vice Chairperson (Technical), IEEE KU SB.

Experience (continued)

- 2020  **Certification.** Technical Support Fundamentals, an online non-credit course authorized by Google and offered through Coursera.
- 2019  **Champion.** Fresher Debaters' League 2019, Khulna University.

References

Reference 01

Abdullah-Al Nahid

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Khulna, Bangladesh.

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Reference 02

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Research Professor

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South Korea.

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